

Projects Options for CESC 510 and/or Senior Design Classes

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8/26/26

Project 1: Small Radio Telescope (SRT) Based on Software Defined Radio (SDR)

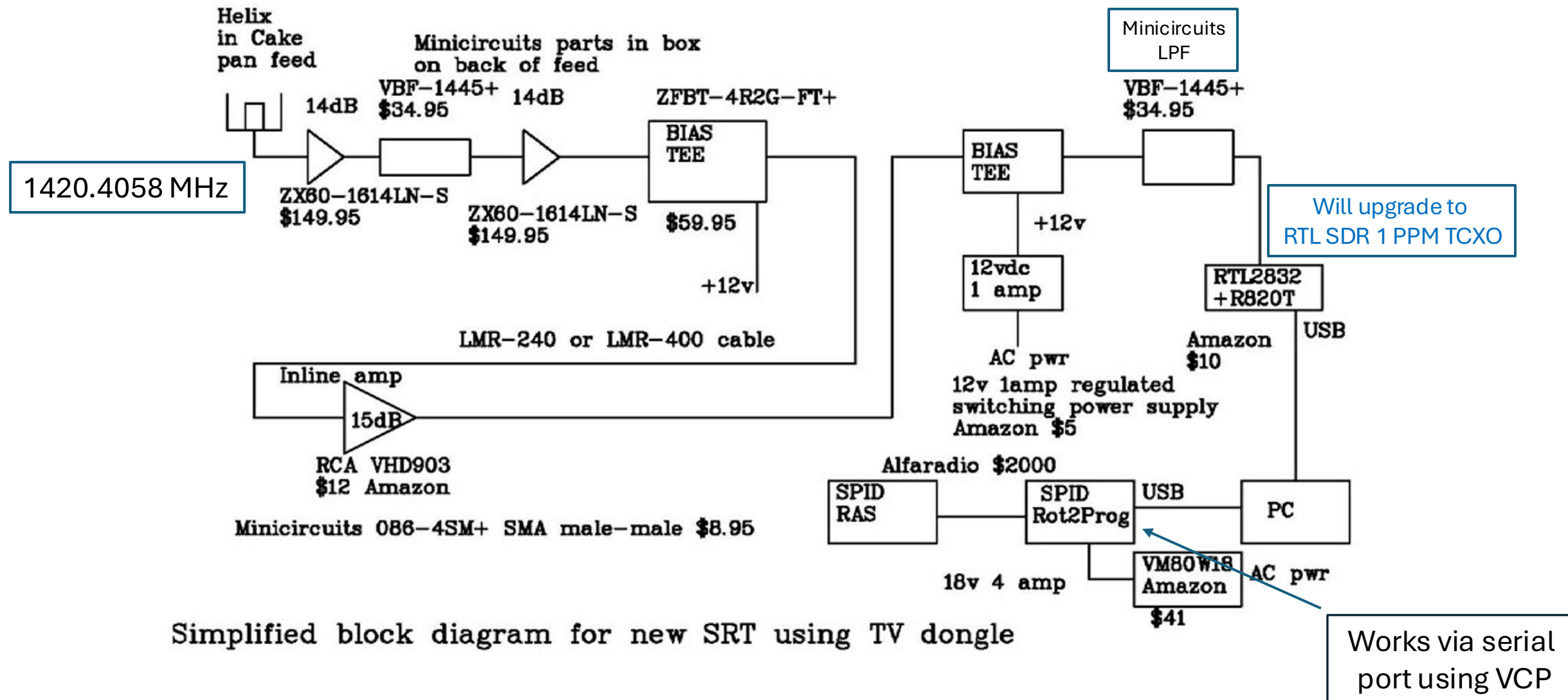
Background info

- Hardware and software were developed by the MIT Haystack Observatory.
 - Home page:
<https://www.haystack.mit.edu/haystack-public-outreach/srt-the-small-radio-telescope-for-education/>
 - GitHub repo for the software:
<https://github.com/MITHaystack/srt-py>
- Installed at Burns Sci Tech at Edge Water.
- Need to upgrade software



Hardware Diagram and Components

Background info



Background info

SPID Azimuth & Elevation antenna rotator **Model: SPID-RAS**

Supplied incl Rot2Prog controller or NEW MD-03 controller



SPID RAS



Rot2Prog Controller

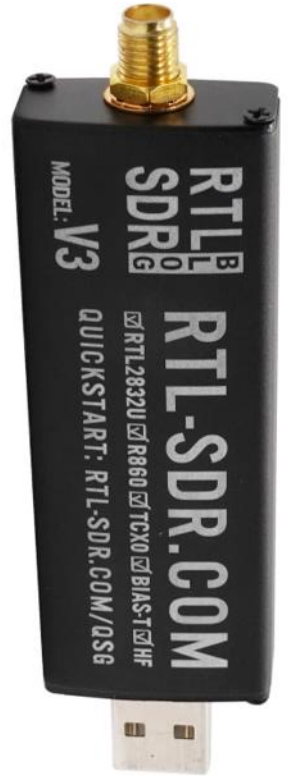


SPID MD-03 Controller

- All above hardware works well
- We will get a new SDR receiver with better frequency stability.

Background info

- **<1 PPM temperature compensated oscillator (TCXO)** – Accurate tuning and almost zero temperature drift (2 PPM max. initial offset, 0.5-1 PPM temperature drift)
- **SMA female antenna port** – Generic dongles use the less common MCX or PAL antenna ports. SMA is more common, so more adapters and antennas are available for it. It is also more durable and has lower RF insertion losses.
- **4.5V USB powered bias tee** - This allows the RTL-SDR to power low noise amplifiers (like the LNA4ALL, HABAMP, RTL-SDR Blog ADS-B LNA) and active antennas through the coax cable. Can be enabled in software (see [V3](#) / [V4](#) users guide).
- **HF Direct Sampling Mode (V3) or HF Upconversion (V4)** - Listen to 500 kHz - 28.8 MHz with direct sampling on the V3 model, or upconversion of the V4 model. There is a built-in 25 MHz low pass filter, but additional HF filters may be required for optimal performance (e.g. to attenuate strong MW AM).
- **Aluminium case and passive cooling** - These units come stock with an aluminium case and passive cooling via a silicon thermal pad. This stops reception failing due to heat when used above ~1.5 GHz.
- **(V4 Models) - Improved front-end filtering**. The V4 model is less prone to being saturated by strong broadcast stations.
- **Improved Antennas** - We provide an excellent dipole antenna starter kit for newbies. Use the included mounts and extension cable to mount the dipole in good position for optimal reception. Receives terrestrial and VHF satellite signals.
- **Various additional improvements compared to other RTL-SDRs** – R820T2 tuner, higher quality passive components, a choke on the USB line to reduce USB noise, a much improved PCB design for significantly less internal spurs and noise, various break-out pads, improved ESD protection, additional bypass capacitors and ferrite power line chokes, improved front end matching circuit, a modified power design for improved long term reliability, modified PLL circuit for better stability at 1.5 GHz and above, and a better LDO.



Software Versions of the SRT

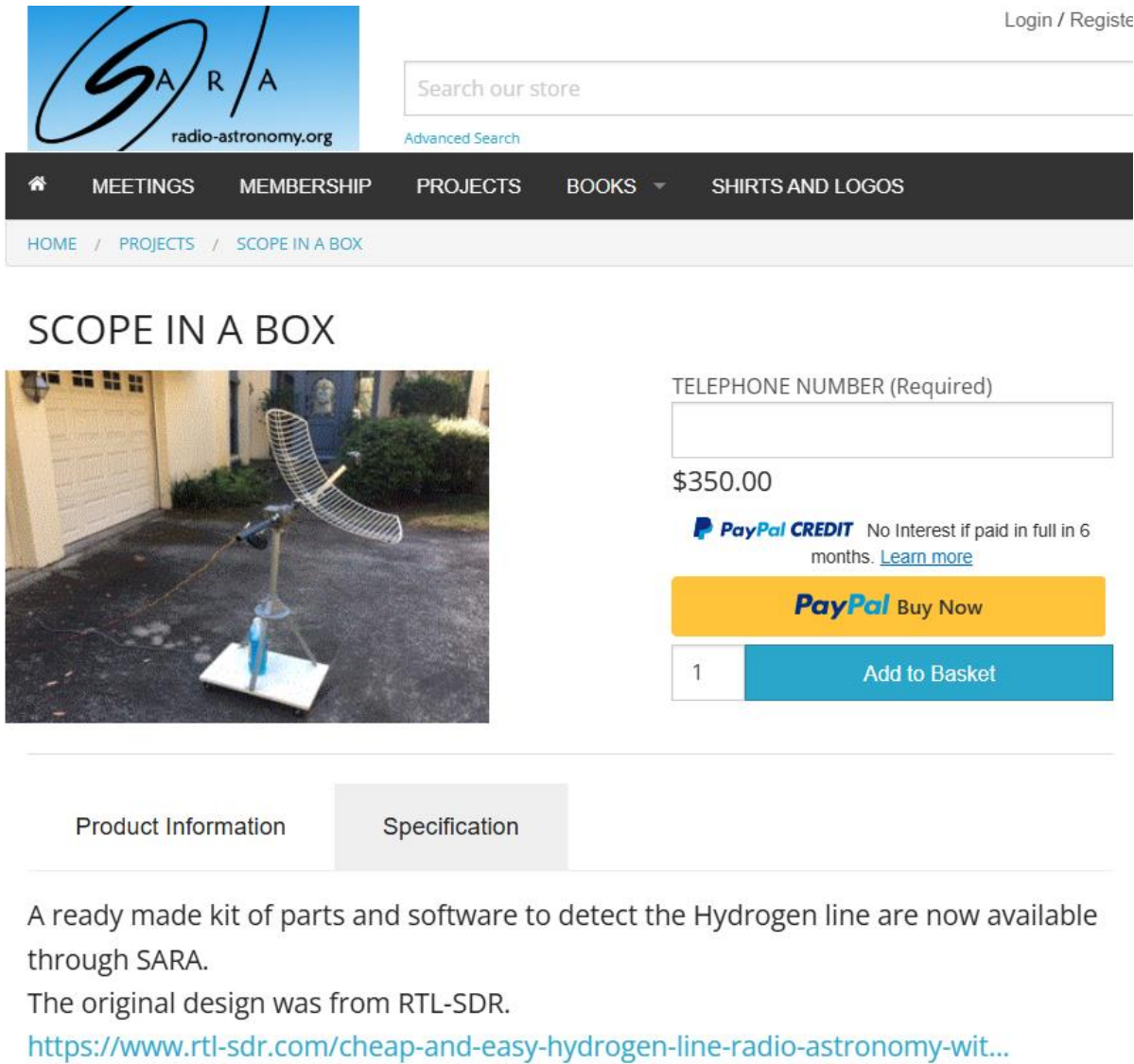
Background info

- There are many versions of SRT software.
 - Long time ago, it was written with Java. See the “Legacy SRT software” section at <https://www.haystack.mit.edu/haystack-public-outreach/srt-the-small-radio-telescope-for-education/#current>.
 - Jeanette Barott and Charlie Husbands were able to make these version work. This was the time when the SRT was installed at the Museum of Art, Science, and History.
 - Then, the programming language is changed to C. See the “Previous SRT software” section at <https://www.haystack.mit.edu/haystack-public-outreach/srt-the-small-radio-telescope-for-education/#current>.
 - The current version is written with Python.

We will switch to a totally different software package, which works well with the current hardware and the new SDR receiver.

Components of the new Software Package

Work to Do



The screenshot shows the website for radio-astronomy.org. The header includes the logo, a search bar, and navigation links. The main content area features a product titled 'SCOPE IN A BOX' with a photograph of the antenna setup. To the right of the image is a purchase form with a telephone number field, a price of \$350.00, a PayPal CREDIT option, a 'PayPal Buy Now' button, and an 'Add to Basket' button. Below the product image are tabs for 'Product Information' and 'Specification'. The 'Product Information' tab is active, displaying text about the kit's purpose and origin, along with a URL.

radio-astronomy.org

Login / Register

Search our store

Advanced Search

MEETINGS MEMBERSHIP PROJECTS BOOKS SHIRTS AND LOGOS

HOME / PROJECTS / SCOPE IN A BOX

SCOPE IN A BOX



TELEPHONE NUMBER (Required)

\$350.00

PayPal CREDIT No Interest if paid in full in 6 months. [Learn more](#)

PayPal Buy Now

1 Add to Basket

Product Information Specification

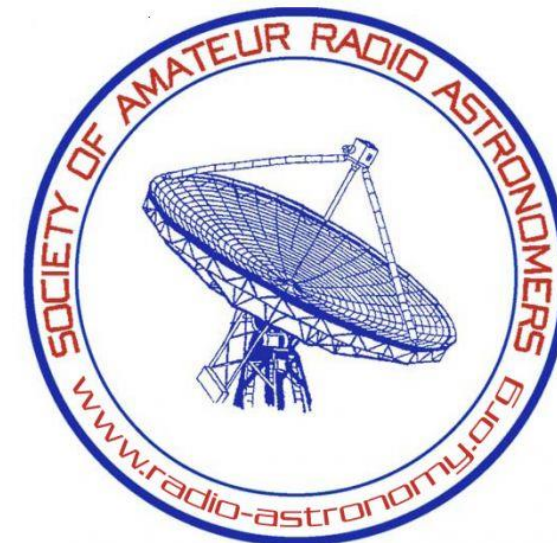
A ready made kit of parts and software to detect the Hydrogen line are now available through SARA.

The original design was from RTL-SDR.

<https://www.rtl-sdr.com/cheap-and-easy-hydrogen-line-radio-astronomy-wit...>

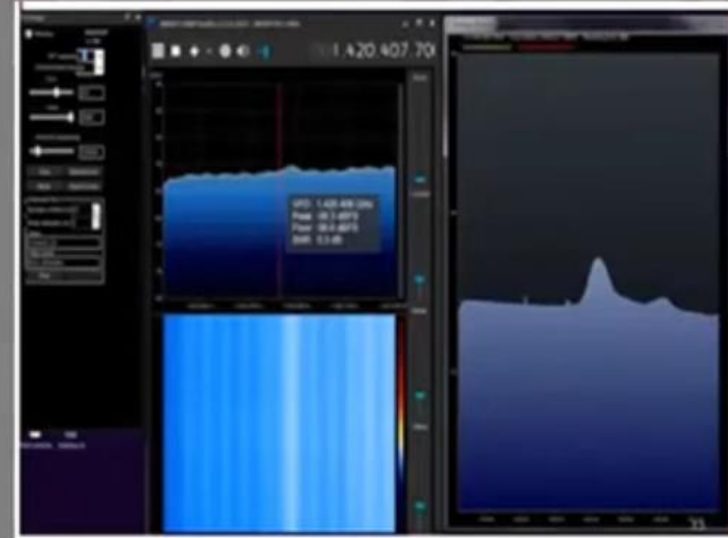
- Processing signals from the receiver.
- Tracking celestial bodies.
- Controlling antenna

Search for amateur radio astronomy software packages



Processing
Signals
from
Receiver

SDR Acquisition :
Air Spy
SDR Sharp Studio
IF Average Plug-In



Astronomical Sky Display
Stellarium



IF_Average Plug-In for AirSpy SDR# Studio

Processing
Signals
from
Receiver



Processing Signals from Receiver



Products ▾

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Download

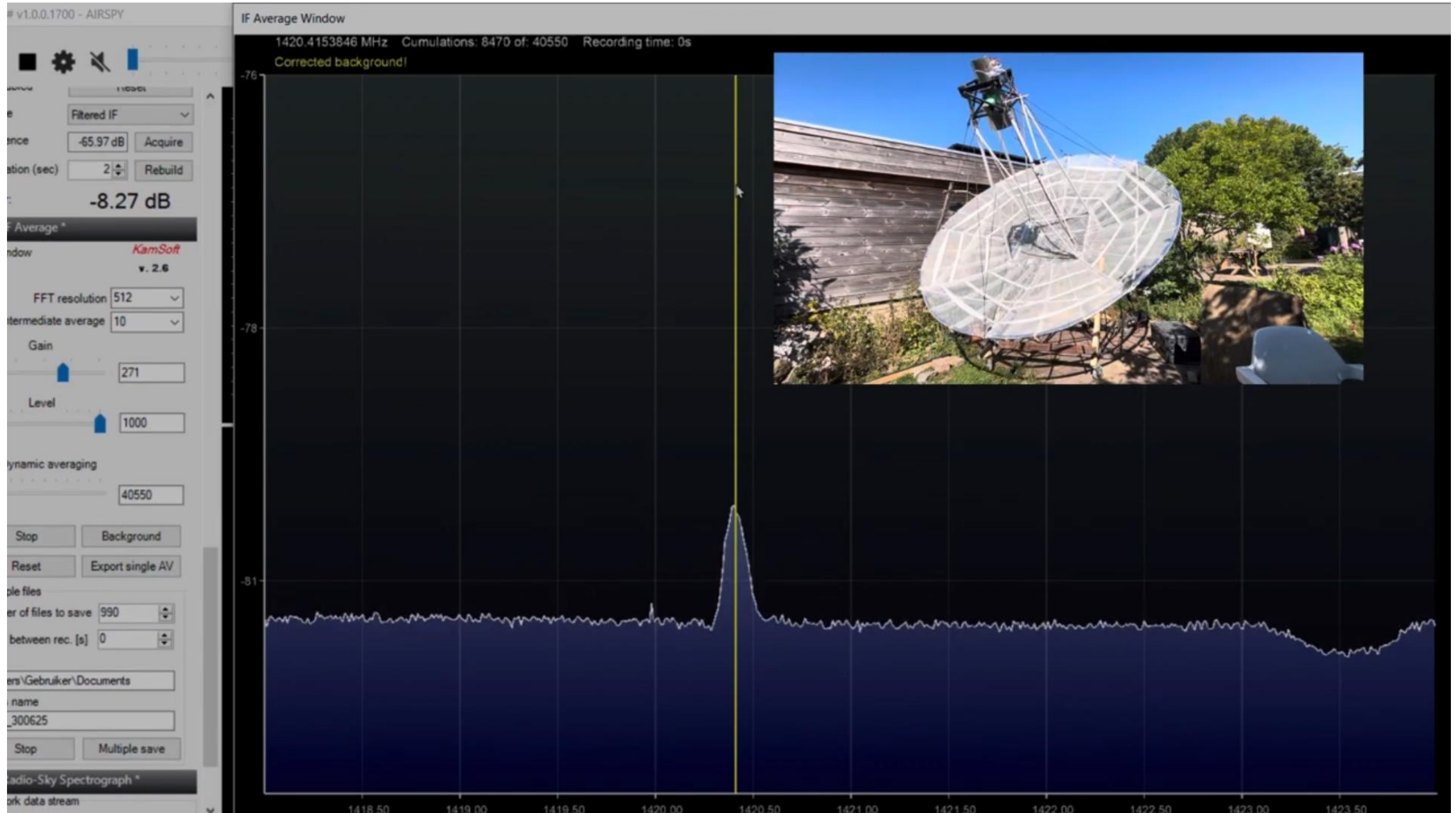
SDR Software Download

Software Defined Radio Package (Change log)

This package contains:

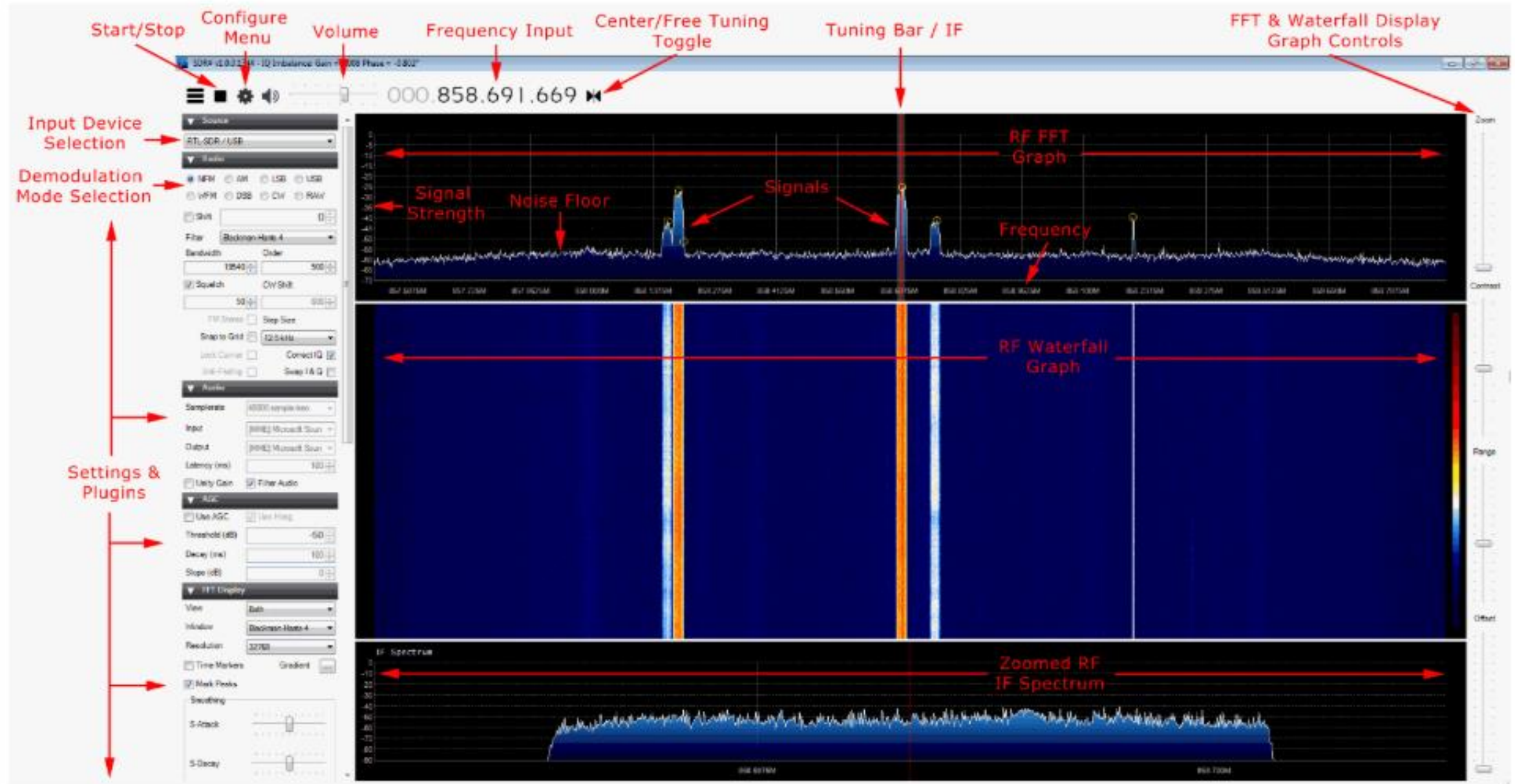
- SDR# (SDRSharp) **revision 1921** (2025-01-01) – The **BEST** free SDR software for Airspy!
- Airspy SDR and SpyServer drivers
- AOR IQ drivers (AR5700D, AR2300IQ, AR5001D with IQ5001 option, AR6000 with IQ5001 option)
- RFSPACE Networked SDR drivers (Net-SDR, SDR-IP, CloudSDR)
- HackRF driver
- RTL-SDR driver installation script (RTL-SDR Blog Build)

Processing Signals from Receiver

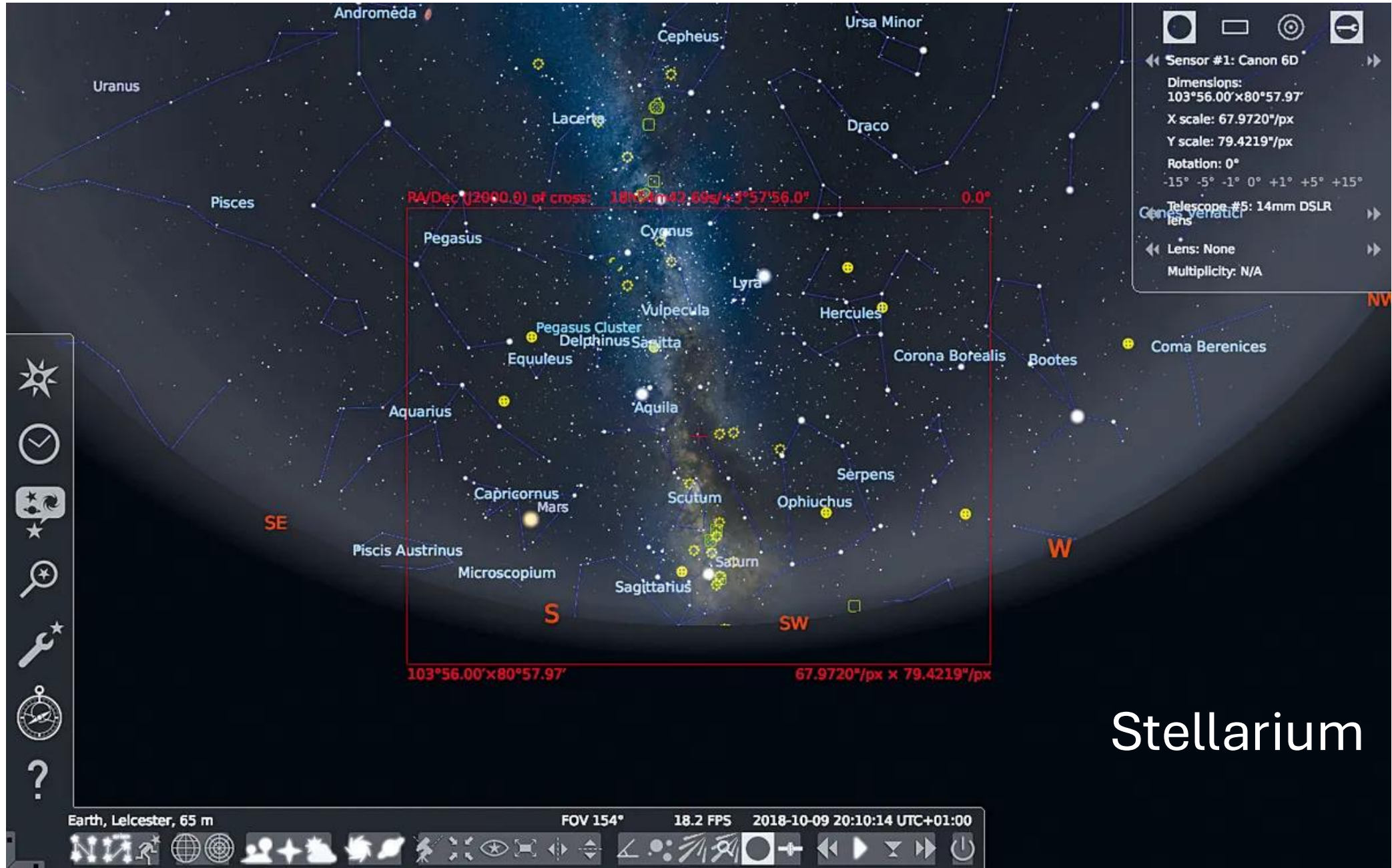


SDR# (SDR Sharp)

Processing Signals from Receiver



Tracking celestial bodies



Stellarium

Controlling Antenna

🏠 / Windows / Utilities & Tools / PstRotator



PstRotator for Windows

Free ★ 4.2 (1 👤) 📥 115

✔️ **Trusted Program** .\ 17.71

Software for
AZ/EL rotators

Free Download
for Windows 

Information Safety scan

Comprehensive Antenna Control for Amateur Radio Operators

PstRotator is a versatile Windows utility designed specifically for amateur radio operators, enabling effective control over antenna rotators for both Azimuth/Elevation and Azimuth-only setups. This free software provides features such as automatic satellite and Earth-Moon-Earth (EME) tracking, Doppler frequency control, and relay switching, making it an essential tool for managing complex antenna systems. Additionally, it supports remote operation through serial ports or network hardware, enhancing flexibility for users.

App specs

- License
Free
- Version
17.71
- Latest update
January 7, 2026
- Platform
Windows
- OS
Windows 11
- Downloads
115 ⓘ

Controlling Antenna

Introduction

PstRotator handles just almost every controllable rotator in common use:

Controller	Protocol	Speed	Web site
Prosistel D	Prosistel	9800	http://www.prosistel.net
Prosistel C	Prosistel	9800	
ERC	DCU-1	4800	http://www.schmidt-alba.de
ERC3D	GS-232A/B	9600	
Green Heron RT21	DCU-1	4800	
Green Heron RT20	DCU-1	4800	
MDS RC-1	DCU-1	4800	
Hy Gain	DCU-1	4800	
Idiom Press Rotor-EZ	DCU-1	4800	
Yaesu GS-232A/B Az	GS-232A/B	9600	
Yaesu GS-232A/B Az/EI	GS-232A/B	9600	
AlfaSpid RAK Az	AlfaSpid	1200	www.rfhamdesign.com
AlfaSpid RAS Az/EI	AlfaSpid (1 deg accuracy)	600	
AlfaSpid RAS Az/EI	AlfaSpid (0.5 deg accuracy)	600	
AlfaSpid BIG-RAS/HR	AlfaSpid (0.1 deg accuracy)	460800	
M2 RC2800	M2	9600	

Goal for Project 1

- Replace the existing software which is not updated regularly.
 - The new software package should be updated by the community regularly.
 - Known issues should be addressed relatively quickly.
- Assemble a relatively easy to use software package for middle school students for science projects.
 - Software package should be very stable so users don't have to face constant operation issues.
 - Appropriate logging is needed

Requirements and Suggestions for Project 1

- For CESC 510:
 - Research what software packages the amateur radio astronomers commonly use which are compatible to our hardware.
 - Install the chosen software package on your computer which runs a SDR for signal receiving and experiment with the basic operations.
 - Do a field test at Burns to make sure the package works.
- For Senior Design:
 - All the above.
 - Create an antenna simulator that can simulate the rotation of the antenna and modulate the strength of received signals.
 - Create a wrapper app so that different pieces of software in the package can be started and operated with the wrapper to reduce the operational overhead and cognitive load.

Project 2: ATC Communication Quality Research Using SDR

Background info

- ATC communications between ATC controller and pilots still are based on classic VHF AM technology.
- Voice quality of ATC communication can vary.
- We want to figure what are the factors that affects the quality of VHF AM communications.
- Result of quality research can be used in two ways:
 - Improve audio quality of the receiver output so that an ASR (automatic speech recognition) system can work better.
 - Generate realistic voice signals that can be used for ASR model finetuning.

Getting ATC Voice Data

- Requesting ATC voice data from LiveATC and other sources, such as FOIA.
- Recording ATC voice data using our own hardware and software:
 - We use the same SDR as Project 1 to create VHF AM receivers.
 - There are many different choices for software to get the audio signal out:
 - Use existing SDR software packages with AM demodulation, such as AirSpy.
 - Create our own AM receiver based on tools like GNU Radio.

Goal/Requirements of Project 2

- For CESC 510: Can select one of TWO flavors:
 - Flavor 1: Voice quality research
 - Analyze requested voice data to document the variation of voice quality for different location, ATC positions, frequencies, and time.
 - Associate the spectrum of the audio signal and the audio quality and investigate root cause for reduced audio quality.
 - Record data by experimenting on slightly changing the receiver frequency and tweaking filter shapes to mimic different voice qualities.
 - Simulate the creation of reduced quality audio signals.

Goal/Requirements of Project 2

- For CESC 510: Can select one of TWO flavors:
 - Flavor 2: Noise characteristic research
 - Analyze noise patterns in the requested voice data.
 - Patterns may include time domain distributions of noise signal strength and well as frequency domain spectrum pattern.
 - Associate the pattern and strength of noise to distance between transmitter and receiver if possible. The distance can be calculated by using ADS-B data.
 - Associate the spectrum of the voice signal and the adjacent noise signal to see the effect of receiver filters.
 - Experiment the creation of noise that have similar noise patterns as that from requested data.

Goal/Requirements of Project 2

- For Senior Design Classes:
 - The union of the above two flavors.

Additional requirements may be added later.